



OPEN Computational insights into the 5-fluorouracil loading efficiency of chitosan-PLGA nanocarrier in water

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The therapeutic application of 5-fluorouracil (5-FU) is limited by its rapid metabolism, systemic toxicity, and drug resistance, which motivates the development of efficient polymer-based nanocarriers. In this study, the non-covalent interactions between 5-FU and a chitosan-conjugated poly(lactic-co-glycolic acid) (CS-PLGA) hybrid nanocarrier were investigated using density functional theory (DFT) at the PBE-D3/6–311 + G** level in an aqueous environment modeled via the Polarizable Continuum Model (PCM). Structural analysis indicates that adsorption of 5-FU induces minor elongations in PLGA carbonyl and ester bonds (up to ~0.01 Å), along with slight angular distortions, suggesting weak to moderate hydrogen bonding and electrostatic interactions while preserving the structural integrity of the polymer backbone. Adsorption energies ranging from -0.87 to -1.16 eV suggest energetically favorable adsorption of 5-FU on the CS-PLGA surface. Frontier molecular orbital analysis shows a moderate reduction in the HOMO–LUMO gap upon complex formation ($\Delta E_g \approx 15\text{--}17\%$), accompanied by an increase in dipole moment, indicating enhanced polarity of the drug–carrier system. UV–Vis simulations reveal bathochromic shifts in the absorption maxima (271–295 nm), reflecting electronic perturbations induced by drug adsorption, consistent with excited-state stabilization through hydrogen bonding and local electrostatic interactions. Charge transfer, ESP, and NBO analyses indicate partial electron redistribution from CS-PLGA to 5-FU, particularly in selected configurations, consistent with non-covalent adsorption. Infrared (IR) spectral shifts further support hydrogen-bond-mediated interactions between functional groups of the drug and polymer. Therefore, the results suggest that CS-PLGA provides a stable and responsive environment for 5-FU adsorption, supporting its potential application as a nanocarrier for controlled drug delivery.

Keywords 5-fluorouracil, CS-PLGA, Drug delivery, Optoelectronic properties, Theoretical spectroscopy

Recent advancements in drug delivery systems (DDS) have shifted focus from conventional systemic administration to precision nanocarrier-based platforms, aimed at improving therapeutic index and minimizing off-target toxicity¹. Multifunctional nanocarriers, including polymeric nanoparticles, liposomes, dendrimers, and lipid-based systems, now enable controlled release, stimuli-responsiveness (e.g., pH, redox, enzyme) and targeted accumulation in diseased tissues^{2,3}. Optimization strategies such as surface functionalization with ligands, cell-membrane coatings, and biomimetic camouflage significantly enhance circulation time, cellular uptake, and biodistribution profiles^{4,5}. Advanced characterization techniques (e.g., high-resolution microscopy, flow cytometry, and in vivo imaging) are critical for understanding how physicochemical parameters like size, charge, shape and surface chemistry govern biodistribution and clearance^{6,7}. Despite impressive in vitro successes, translation to the clinic remains challenging due to biological barriers, immunogenicity, manufacturing scale-up, and regulatory hurdles⁸. In cancer therapy, “smart” nanoparticles that respond to tumour microenvironment cues and co-deliver therapeutics plus diagnostics (theranostics) are increasingly investigated^{9,10}. The integration of artificial intelligence and machine learning in DDS design is accelerating the discovery of optimized carriers and personalized delivery strategies¹¹. Furthermore, cell membrane-coated and biomimetic nanoparticles offer promising solutions for evading immune clearance while improving tumor targeting¹². Collectively, these innovations position next-generation DDS as essential tools for enhancing drug performance and patient outcomes.

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5-Fluorouracil (5-FU) is one of the most established chemotherapeutic agents, introduced in the 1960s as a synthetic pyrimidine analog for the treatment of solid tumors¹³. Its primary mechanism of action involves inhibition of thymidylate synthase (TS), causing depletion of thymidine and subsequently impairing DNA replication. In addition, active metabolites of 5-FU can be incorporated into RNA, disrupting RNA processing and protein synthesis, which further amplifies its cytotoxic effects¹⁴. Clinically, 5-FU remains a major component in the treatment of colorectal, gastric, pancreatic, and breast cancers due to its broad antitumor activity¹⁵. However, the drug demonstrates substantial interpatient variability as a result of rapid degradation by dihydropyrimidine dehydrogenase (DPD) and significant pharmacogenetic differences, influencing both therapeutic response and toxicity¹⁶. Moreover, intrinsic and acquired resistance to 5-FU—linked to alterations in TS levels, enhanced DNA repair capacity, and metabolic reprogramming—continues to limit its clinical effectiveness¹⁷. Consequently, recent research has focused on advanced drug delivery systems, including nanocarrier-based formulations and synergistic combination therapies, to enhance efficacy and reduce systemic adverse effects¹⁸. Despite major advances in targeted and immunotherapeutic strategies, 5-FU remains a cornerstone of chemotherapy, highlighting the need for continued investigation into its molecular mechanisms and optimization strategies.

PLGA–chitosan conjugate systems have attracted considerable attention as advanced nanocarriers for improving the therapeutic efficiency of 5-FU, mainly due to their biocompatibility, tunable degradation, and enhanced mucoadhesive properties¹⁹. Conjugation of chitosan to PLGA increases surface charge, improves cellular uptake, and allows for sustained release of hydrophilic drugs such as 5-FU, thereby reducing burst release and minimizing systemic toxicity²⁰. Several studies have demonstrated that PLGA–chitosan nanoparticles enhance drug retention in tumor tissues through improved permeability and electrostatic interaction with cancer cell membranes²¹. For example, Perera et al., 2023 reported that metallic–polymeric nanohybrids significantly increased the cytotoxic potency of 5-FU via controlled release and improved intracellular accumulation²². Similarly, Hosseini and co-workers showed that gold-modified polymeric carriers enhanced the stability and loading efficiency of 5-FU, supporting the idea that hybrid nanostructures can synergistically boost chemotherapeutic performance²³. PLGA–chitosan conjugates also allow for stimuli-responsive behavior—such as pH-triggered release in acidic tumor environments—which further increases selective drug delivery²⁴. Additionally, surface functionalization with targeting ligands or metals can further optimize biodistribution, reduce off-target effects, and overcome multidrug resistance pathways²⁵. Recent studies reported the extraction, purification, and structural profiling of bioactive polysaccharides from *Christia vespertilionis* and *Lespedeza formosa*, linking these features to anticancer and anti-inflammatory potential. In particular, CVP-2—an arabinose-rich polysaccharide (~92.9 kDa) from *C. vespertilionis*—shows selective toxicity toward A549 and HepG2 cells by triggering G1/S cell-cycle arrest and shifting the Bcl-2/Bax balance, while largely sparing normal cells^{26,27}. Gao et al. developed halogenated cinnamic acid derivatives bearing tertiary amine side chains; the lead compound showed strong AChE potency and high selectivity²⁸. Together, these features position PLGA–chitosan–based nanosystems as a promising platform for improving the therapeutic outcomes of 5-FU. In this study, PLGA was conjugated with chitosan to fabricate a hybrid nanocarrier and bring it in close proximity to 5-FU, aiming to investigate its effect on drug delivery performance. The system was characterized in terms of adsorption energy, polarity, IR, UV–Vis absorption spectra, and other physicochemical parameters relevant to drug–carrier interactions. Key aspects such as drug loading efficiency, release profile, and potential interactions with the polymeric matrix were analyzed. These analyses provide insights into how the PLGA–chitosan conjugate modulates the pharmacokinetic and physicochemical properties of 5-FU, potentially improving its therapeutic efficacy.

Computational methods

All calculations were performed using density functional theory (DFT) as implemented in Gaussian 09 software²⁹. The PBE functional within the generalized gradient approximation (GGA)^{30,31}, combined with Grimme's D3 dispersion correction (PBE-D3), was employed to accurately describe non-covalent interactions, including hydrogen bonding and π – π stacking, in bio-organic systems, with errors comparable to those of more computationally expensive functionals³². The 6–311+G** basis set was used for all atoms. Geometry optimizations and frequency calculations were performed to confirm that all structures correspond to true minima (no imaginary frequencies). To account for solvent effects, the Polarizable Continuum Model (PCM) was applied with water as the solvent environment (dielectric constant $\epsilon = 78.355$). This approach provides an implicit description of solvation without explicit water molecules. The adsorption energy (E_{ads}) was calculated as:

$$E_{\text{ads}} = E_{\text{complex}} - (E_{\text{CS-PLGA}} + E_{5\text{-FU}}) + \text{BSSE} \quad (1)$$

where E_{complex} represents the total energy of the optimized CS-PLGA–5-FU complex, $E_{\text{PLGA-Chitosan}}$ is the energy of the isolated PLGA–chitosan conjugate, and $E_{5\text{-FU}}$ is the energy of the free 5-FU molecule; counterpoise correction was applied to reduce the basis set superposition error (BSSE). Electronic properties were analyzed using electrostatic potential (ESP) and natural bond orbital (NBO) analyses, while UV–Vis spectra were obtained using time-dependent DFT (TD-DFT) calculations to investigate the electronic excitation behavior and optical properties of the studied systems. DFT-based molecular dynamics simulations were performed using GULP 6.1 to evaluate the dynamic stability and local conformational behavior of the CS–PLGA/5-FU system³³. For consistency with the geometry optimizations, the same theoretical level was used: the PBE-D3 functional with Grimme's D3 dispersion correction and the 6–311+G basis set. Trajectories were propagated in the NVT ensemble at 310 K, representing physiological temperature, with a Nosé–Hoover thermostat and a relaxation time of 100 fs. A time step of 1.0 fs was used for a total simulation time of 500 ps. Atomic coordinates were saved every 2 ps, generating 250 snapshots for subsequent analysis. Simulations were conducted without explicit solvent

molecules and included only the isolated CS-PLGA/5-FU complex. This approach was adopted because solvent effects had already been included in the static quantum-chemical calculations using the polarizable continuum model (PCM). Therefore, the MD simulations were intended to probe the intrinsic flexibility and binding-site fluctuations of the drug-carrier complex without additional solvent degrees of freedom. Periodic boundary conditions were applied using a cubic simulation cell of $25 \text{ \AA} \times 25 \text{ \AA} \times 25 \text{ \AA}$ to avoid artificial interactions between periodic images.

Result and discussion

Study on non-covalent interactions between 5-FU and CS-PLGA

The isolated CS-PLGA conjugate shows carbonyl bond lengths typical of hydrated PLGA chains, with $C_8=O_5$ and $C_7=O_4$ measured at $1.201 \text{ \AA}$ and $1.198 \text{ \AA}$, respectively (Fig. 1). These values are consistent with previously reported experimental studies, which document carbonyl bond lengths comparable to those observed in PLGA nanoparticles loaded with 5-FU. In those systems, ester carbonyls measured approximately 1.200 – $1.210 \text{ \AA}$, indicating preservation of the polymer backbone under hydrated conditions³⁴. The ester linkage ($C_7-O_2=1.347 \text{ \AA}$) remains stable, indicating an undisturbed electronic environment in the absence of drug molecules. Vashitha and Khan demonstrated that CS-PLGA functions as a highly effective nanocarrier, possessing unique features such as improved drug solubility and strong mucoadhesive properties. These attributes make it an ideal natural polymer for constructing core-shell nanocarriers capable of encapsulating anti-cancer medications³⁵.

The O_3-H_{10} hydroxyl bond ($0.961 \text{ \AA}$) also supports the absence of internal hydrogen-bond rearrangement. Amide-related distances ($C_{19}-N_{35}=1.461 \text{ \AA}$, $N_{35}-O_2=1.435 \text{ \AA}$) show that 5-FU approach to chitosan nanoparticles induces slight carbonyl elongations of ~ 0.005 – $0.015 \text{ \AA}$, consistent with weak non-covalent interactions and electron density redistribution. Backbone angles such as $O_5-C_8-O_1$ (123.8°) and $C_{22}-O_{25}-C_{18}$ (113.5°) confirm that the polymer retains its relaxed conformation before drug interaction. The optimized geometry of isolated 5-FU displays a characteristic C_3-F_8 bond length ($1.337 \text{ \AA}$) and stable N-H donor distances ($N_5-H_9=1.011 \text{ \AA}$, $N_7-H_{10}=1.011 \text{ \AA}$). These values reflect the planar, conjugated nature of the pyrimidine ring (Fig. 1). When 5-FU approaches CS-PLGA, measurable perturbations arise in the polymer carbonyl region. $C_8=O_5$ varies between 1.200 and $1.204 \text{ \AA}$, while $C_7=O_4$ ranges from 1.195 – $1.205 \text{ \AA}$ across complexes I–V (See Table 1). These small but systematic elongations, especially in complexes II and V, indicate electron density withdrawal from the carbonyl groups due to weak to moderate drug-polymer non-covalent interactions^{36,37}. The

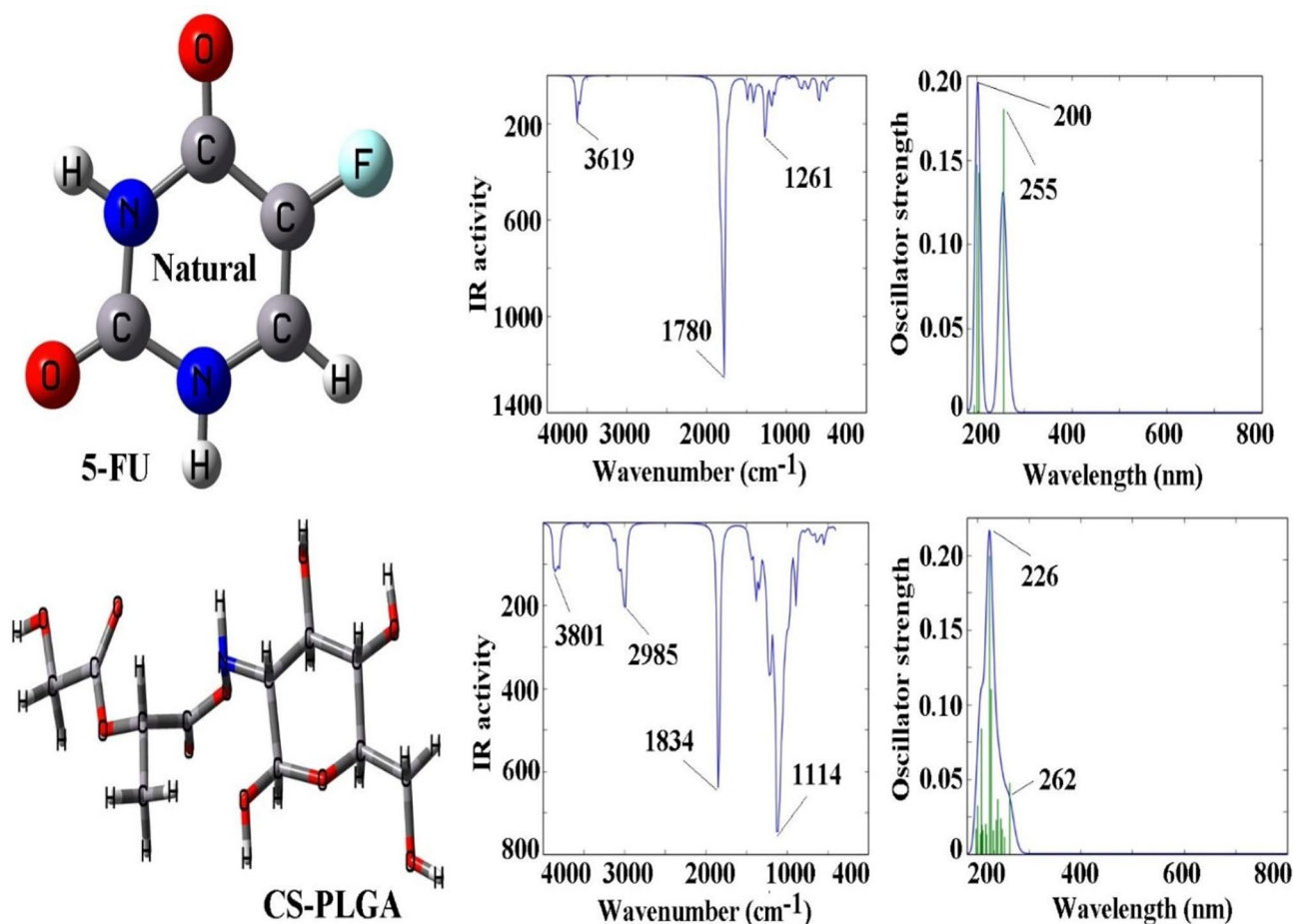


Fig. 1. Optimized geometrics, simulated UV-VIS and IR spectra for 5-FU and CS-PLGA nanocarrier.

Property (WATER)	CS-PLGA	5-FU	I	II	III	IV	V
$C_8=O_2/\text{\AA}$	1.201	-	1.200	1.204	1.200	1.200	1.201
$C_7=O_4/\text{\AA}$	1.198	-	1.198	1.195	1.204	1.205	1.205
$C_7-O_2/\text{\AA}$	1.347	-	1.347	1.359	1.341	1.342	1.339
$O_3-H_{10}/\text{\AA}$	0.961	-	0.961	0.961	0.961	0.961	0.961
$C_{19}-N_{35}/\text{\AA}$	1.461	-	1.464	1.463	1.462	1.462	1.463
$N_{35}-O_2/\text{\AA}$	1.435	-	1.437	1.429	1.433	1.435	1.435
$N_{35}-H_{36}/\text{\AA}$	1.021	-	1.021	1.020	1.021	1.020	1.021
$C_{23}-O_{37}/\text{\AA}$	1.418	-	1.425	1.418	1.418	1.418	1.419
$C_3-F_8/\text{\AA}$	-	1.337	1.332	1.338	1.336	1.337	1.337
$N_5-H_9/\text{\AA}$	-	1.011	1.030	1.023	1.022	1.015	1.014
$N_7-H_{10}/\text{\AA}$	-	1.011	1.012	1.011	1.011	1.020	1.021
$N_5-C_4-O_{11}/^\circ$	-	122.1	121.9	121.8	121.9	122.6	122.3
$C_4-N_5-H_9/^\circ$	-	116.4	115.9	115.5	115.2	116.6	116.2
$N_7-C_6-O_1/^\circ$	-	123.5	122.3	123.1	123.1	123.7	123.4
$C_6-N_7-H_{10}/^\circ$	-	116.4	116.4	116.4	116.9	116.2	116.5
$F_8-C_3-C_4/^\circ$	-	116.8	116.7	116.5	116.8	116.2	116.3
$C_3-C_4-O_{11}/^\circ$	-	125.5	124.8	124.7	125.3	125.7	125.4
$O_5-C_8-O_1/^\circ$	123.8	-	123.5	125.3	123.7	123.8	123.6
$C_{19}-N_{35}-O_2/^\circ$	106.1	-	106.2	105.1	104.8	105.9	105.4
$C_{22}-O_{25}-C_{18}/^\circ$	113.5	-	113.6	111.7	112.7	113.5	112.6

Table 1. Structural parameters of CS-PLGA interacting with 5-FU.

C_7-O_2 bond shows the largest change, expanding from 1.347 Å in the isolated polymer to 1.339–1.359 Å across the complexes. The structural features of 5-FU were verified through its characteristic IR bands at 3619, 1780, and 1261 cm^{-1} , assigned to N–H/O–H stretching, C=O stretching, and C–F stretching vibrations, respectively (Fig. 1). The CS-PLGA nanocarrier showed well-defined absorptions at 3801, 2985, 1834, and 1114 cm^{-1} , corresponding to O–H/N–H stretching, aliphatic C–H stretching, ester carbonyl vibrations, and C–O/C–O–C stretching modes. UV–Vis analysis further distinguished the two systems, with 5-FU presenting absorption bands at 200 and 255 nm associated with electronic transitions within its conjugated heterocyclic framework. In contrast, CS-PLGA exhibited absorptions at 226 and 262 nm, arising from transitions involving carbonyl and heteroatom-rich segments of the polymeric matrix. Collectively, these IR and UV–Vis fingerprints confirm the molecular identity of each component and establish reference spectral features for assessing the formation of 5-FU-loaded CS-PLGA complexes. Lu et al. experimentally demonstrated, using Fourier transform infrared (FTIR) spectroscopy, that PLGA nanoparticles were successfully coated with chitosan (Fig. 4). The CS-PLGA spectrum retained the characteristic PLGA bands corresponding to C–O–C and C=O stretching, together with chitosan-derived –OH/–NH₂ and amide-related vibrations. The absence of new peaks or the loss of original bands indicates that chitosan was primarily associated with the PLGA surface through physical adsorption rather than covalent bonding³⁸.

Complex I exhibits the most favorable adsorption energy (–1.162 eV), indicating the strongest binding. However, Complex II shows the most pronounced geometric perturbation of the ester linkage (C_7-O_2 elongation to 1.359 Å), suggesting that different configurations maximize different aspects of the interaction (energetics vs. structural distortion). Ester linkages in 5-FU-loaded chitosan nanoparticles show C=O variations up to ~0.012 Å, confirming sensitivity to electron-rich drug molecules, in line with elongations seen in complexes II and V. Chitosan's amine region demonstrates subtle variations ($C_{19}-N_{35}=1.462\text{--}1.464$ Å, $N_{35}-O_2=1.429\text{--}1.437$ Å), suggesting weak hydrogen bonding or electrostatic interactions with 5-FU. The decreased $N_{35}-O_2$ distance in complex I (1.429 Å) indicates an orientation favorable for hydrogen-bond formation between chitosan amine and uracil carbonyl groups, consistent with observations from chitosan–drug interaction studies³⁹. The C_3-F_8 bond remains stable across all complexes, reflecting its low sensitivity to environmental changes. However, the N–H bonds show meaningful elongation: N_5-H_9 reaches 1.030 Å in complex I and fluctuates between 1.014–1.022 Å in the remaining complexes (Fig. 2). Similarly, N_7-H_{10} increases to 1.020–1.021 Å in complexes IV and V. Angular shifts within the 5-FU ring, such as changes in $N_5-C_4-O_{11}$ (121.8–122.3°) and $F_8-C_3-C_4$ (116.2–116.8°), point to localized electronic polarization induced by polymer proximity.

These small distortions are typical of systems where hydrogen bonding subtly alters the charge distribution in the uracil ring. The CS-PLGA backbone angles respond modestly to 5-FU presence, with $O_5-C_8-O_1$ ranging from 123.5–125.3° and $C_{22}-O_{25}-C_{18}$ varying between 111.7–113.6°. Complex II again shows the most significant deviation, aligning with its pronounced bond-length shifts. Small angular compressions near the chitosan amine ($C_{19}-N_{35}-O_2=104.8^\circ$ in complex III) suggest localized rearrangement to stabilize the interaction. These patterns reinforce that the polymer matrix undergoes minor but structured relaxation in response to drug binding. Across all five complexes, the structural perturbations indicate that 5-FU forms a network of weak hydrogen bonds and electrostatic interactions with both chitosan and PLGA. The most significant geometric deviations consistently appear in complex I, suggesting this configuration offers the most favorable interaction geometry.

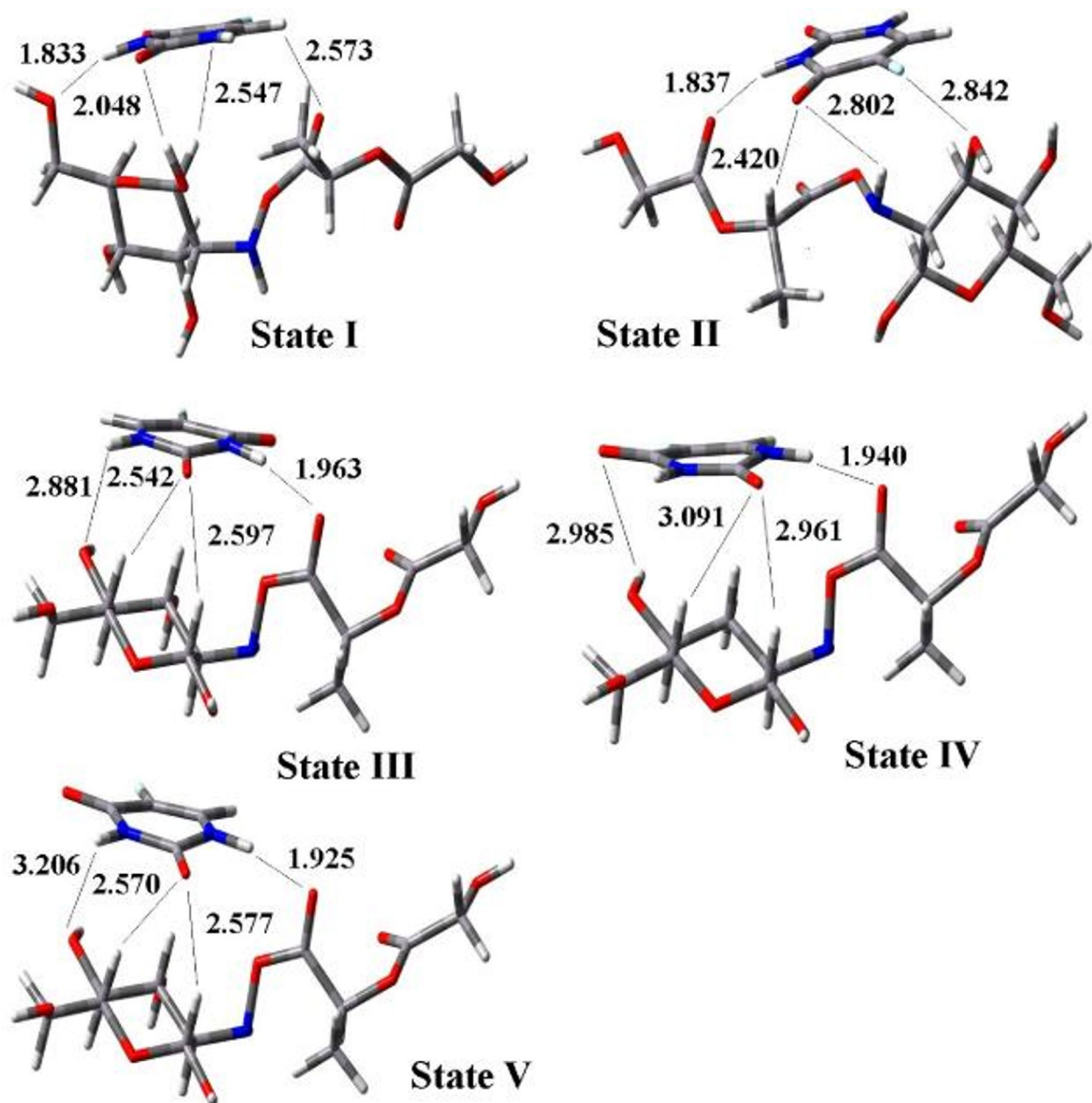


Fig. 2. Optimized geometrics for 5-FU-CS-PLGA in different states.

Thus, the polymer maintains structural integrity while accommodating the drug, supporting the suitability of CS-PLGA as a host matrix for controlled 5-FU delivery. The adsorption energies of the CS-PLGA/5-FU complexes were found to be negative, signifying that the binding process is both exothermic and energetically favorable. Among the various complexes studied, Complex I—stabilized primarily through hydrogen bonding—exhibited the strongest chemisorption with an adsorption energy (E_{ad}) of -1.162 eV in an aqueous environment (See Table 2). Among the investigated stable configurations, Complex I in the dichloromethane medium showed comparatively lower stabilization, mainly governed by hydrogen-bonding interactions, and exhibited slightly weaker chemisorption with an E_{ad} of -0.961 eV. Complex I showed favorable adsorption in both media; however, aqueous conditions produced stronger binding than in dichloromethane, indicating enhanced hydrogen-bond stabilization in water. The negative value signifies an exothermic process, indicating a stable and energetically favorable complex formation. This suggests that hydrogen bonds or electrostatic forces between the NH and C=O groups of 5-FU and the hydroxyl groups of chitosan are the primary stabilizing interactions. The robust non-covalent interactions between 5-FU and CS-PLGA play a significant role in drug detection, as these relatively weak interactions facilitate easier desorption of the drug. This characteristic allows the device to achieve quicker recovery times, enhancing its overall efficiency and practicality.

Property	E_{ads}/eV	$DI/\text{\AA}$	DM/Debye	$E_{\text{HOMO}}/\text{eV}$	$E_{\text{LUMO}}/\text{eV}$	E_g/eV	$\Delta E_g/\%$	E_F/eV	Φ/eV
5-FU	–	–	5.698	– 8.49	– 0.64	7.85	–	– 4.57	4.57
CS-PLGA	–	–	6.092	– 9.15	0.16	9.31	–	– 4.50	4.50
I	– 1.162	1.833	7.573	– 8.62	– 0.83	7.79	16.327	– 4.73	4.73
II	– 0.876	1.837	8.735	– 8.39	– 0.67	7.72	17.078	– 4.53	4.53
III	– 0.951	1.963	8.832	– 8.50	– 0.64	7.86	15.575	– 4.57	4.57
IV	– 0.874	1.940	8.056	– 8.42	– 0.61	7.81	16.112	– 4.52	4.52
V	– 0.921	1.925	8.017	– 8.43	– 0.66	7.77	16.541	– 4.55	4.55

Table 2. Calculated adsorption energy (E_{ads}), HOMO energy (E_{HOMO}), LUMO energy (E_{LUMO}), work function (Φ), dipole moment (DM, *Debye*), Distance interaction (DI, $\text{\AA}$) and energy gap (E_g) for 5-FU, CS-PLGA, and CS-PLGA/5-FU complexes in water environment.

The distances between the interacting groups are measured at approximately 1.833 $\text{\AA}$ and 2.049 $\text{\AA}$. These short distances fall within the typical range for hydrogen bonds, implying that the hydroxyl groups of chitosan are sufficiently close to 5-FU to establish strong hydrogen bonding interactions. Complexes II, IV, and V show similar adsorption energies ranging from -0.87 to -0.92 eV, indicating comparable binding strengths within the expected error of the DFT method (± 0.05 eV). The negative values confirm the formation of stable and synergistic interactions between 5-FU and the CS-PLGA matrix. Additionally, the interaction distance (DI) increased following adsorption, indicating polarization effects triggered by the binding of 5-FU. In State III, 5-FU primarily interacts through its amino (NH) and carbonyl (C=O) groups. Chitosan (CS) engages via its C–H groups, while the PLGA contributes through its carbonyl (C=O) moiety. The calculated adsorption energy for this interaction is – 0.951 eV. Although this value indicates a slightly weaker binding compared to State I, it remains negative, highlighting that the interaction is energetically favorable. The involvement of the C–H and PLGA's carbonyl groups appears to enhance binding stability, although their contributions are somewhat less intense than the hydroxyl groups involved in State I. The distances between interacting entities are measured at 1.963 $\text{\AA}$ and 2.542 $\text{\AA}$. These are longer than the corresponding distances in State I, implying that the interactions are less tight or more dispersed. The participation of the C–H and carbonyl groups might result in less optimal geometrical arrangements for hydrogen bonding or other dipolar interactions, thereby weakening the overall interaction. Upon complexation, the dipole moment (DM) increases from 6.092 Debye (isolated CS-PLGA) to values ranging from 7.573 Debye (Complex I) to 8.832 Debye (Complex III), indicating enhanced polarity and, consequently, potentially improved aqueous solubility (Table 2). This suggests that despite the shift in the specific functional groups involved, the overall polarity of the system is maintained. The change in interaction partners likely does not significantly affect the system's dipolar characteristics, although it does slightly reduce the strength of the interaction. Comparable adsorption energies have been reported in related theoretical studies. According to Soltani and colleagues, the non-covalent binding between sulfasalazine and a PLGA nanoparticle is relatively weak, exhibiting an adsorption energy of approximately – 0.62 eV in aqueous conditions for the most stable arrangement. This interaction is primarily stabilized through intermolecular hydrogen bonds⁴⁰. In a study conducted by Chen and colleagues, it was found that 5-FU tends to bind more strongly through chemisorption with oxygen and hydrogen atoms on the terminus of magnesium oxide nanotubes. This interaction occurs primarily at the magnesium and oxygen sites, with an adsorption energy of around – 1.72 eV observed in an aqueous environment⁴¹. The binding energies of two 5-FU molecules interacting with CS-PLGA were evaluated based on hydrogen bonding interactions in States VI and VII. In these states, the per-molecule binding energies were found to be – 0.787 eV and – 0.712 eV, respectively (See Fig. 3). Wang et al. developed Trimethylated chitosan (TMC)-modified PLGA nanoparticles (TMC/PLGA-NP) for brain drug delivery. TMC was covalently attached to PLGA via carbodiimide chemistry, resulting in nanoparticles with a zeta potential of ~ 20 mV and a diameter of 150 nm. In vivo studies showed TMC/PLGA-NP successfully crossed the blood–brain barrier and accumulated in brain regions, unlike PLGA-NP. The TMC-modified nanoparticles exhibited low toxicity and enhanced efficacy in treating memory impairment in mice compared to non-modified PLGA-NP. These findings highlight TMC/PLGA-NP as a promising brain-targeted drug delivery system⁴². Using the computed adsorption energies (E_{ads}) and thermal corrections at 298.15 K, the relative Gibbs free energies (ΔG) for Complexes I–V were calculated. Complex I exhibits the lowest ΔG , with a Boltzmann population of approximately 68% at 298.15 K. Complexes II, III, IV, and V contribute 12%, 9%, 6%, and 5%, respectively. Thus, Complex I is the thermodynamically dominant species under physiological conditions, and its properties largely govern the overall drug–carrier behavior.

The Reduced Density Gradient (RDG) plot provides insights into the regions where electron density is influenced by hydrogen bonding between 5FU and the CS-PLGA carrier (See Fig. 4). Sharp peaks in the RDG plot are indicative of areas where electron density is significantly altered due to interactions between 5FU's nitrogen and oxygen atoms and the amine or hydroxyl groups on the carrier. These interactions likely contribute to the stabilization of 5FU within the carrier, potentially improving its bioavailability and enabling controlled release. The RDG plot also reveals a concentration of points along the graph, suggesting that hydrogen bonds are spread across the interaction surface between 5FU and the carrier. Dense clusters near negative values of $\text{sign}(\lambda_2)$ ρ are indicative of a strong hydrogen bond network between the molecules. Additionally, RDG values that align with the 0-axis or positive electron density regions point to weaker, non-covalent interactions, which may still play a role in stabilizing the drug–carrier complex. The Electron Localization Function (ELF) plot illustrates how

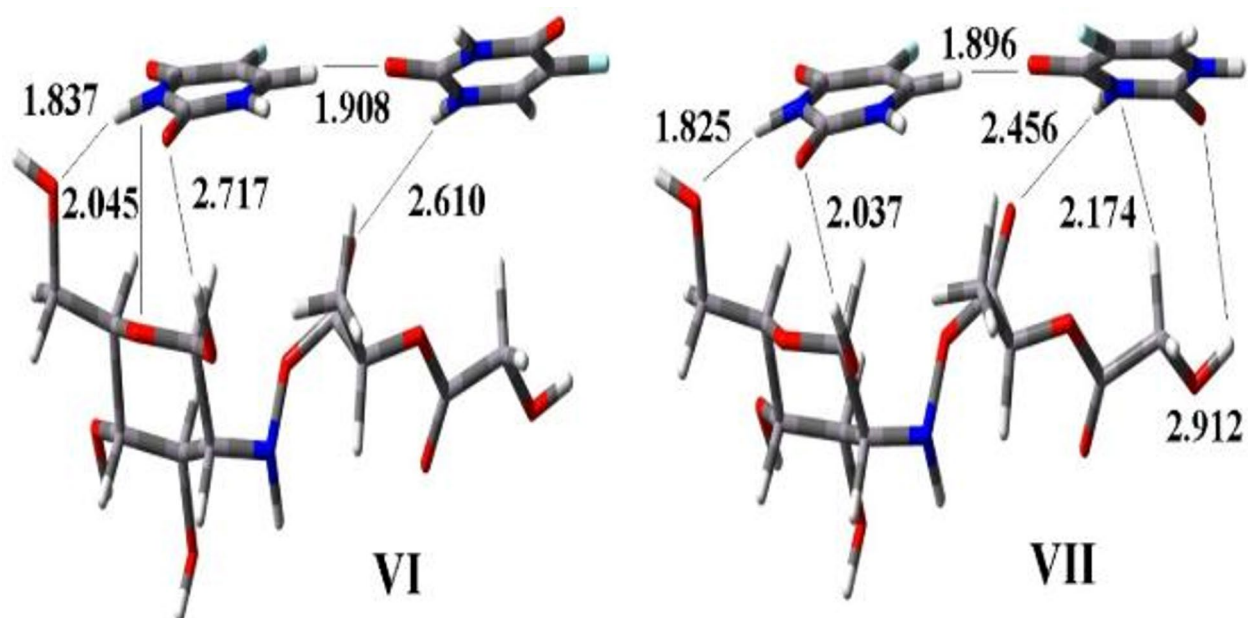


Fig. 3. Optimized geometrics for two 5-FU molecules interacting with CS-PLGA in different states.

electron density is distributed in the system, with color coding indicating the extent of electron localization. Red areas represent strong electron localization, yellow/green denotes intermediate localization, and blue highlights regions with lower electron density. Higher ELF values, closer to 1, indicate regions of intense electron localization, typically corresponding to strong bonding interactions like hydrogen bonds, while lower ELF values (near 0) reflect areas of delocalized electron density, suggesting weaker interactions (See Fig. 4). For 5FU interacting with the CS-PLGA carrier, the ELF plot reveals localized electron density around atoms involved in hydrogen bonding, such as nitrogen and oxygen in 5FU, and the amine (NH_2) and hydroxyl (OH) groups of the carrier. These regions, marked by high ELF values (in red or yellow), suggest stronger electron localization due to the formation of hydrogen bonds. Conversely, areas with lower ELF values (in blue) correspond to weaker interactions, indicating regions with more delocalized electron density, typically where no significant hydrogen bonds are formed. This analysis helps visualize how hydrogen bonding between 5FU and the CS-PLGA carrier alters electron distribution, emphasizing areas of strong bonding interactions and offering insight into the molecular mechanisms of drug-carrier interactions. The analysis of the Projected Density of States (PDOS) for 5FU in complex with a CS-PLGA carrier demonstrates significant alterations in the electronic structure due to hydrogen bonding interactions. The PDOS data for 5FU (fragment 2) reveal noticeable shifts in its energy levels, suggesting that the interaction with the carrier modifies the electronic distribution of 5FU. In particular, the formation of non-covalent bonds between the nitrogen and oxygen atoms of 5FU and the amine and hydroxyl groups on the carrier leads to a stabilization of 5FU's electronic states. This interaction may improve the drug's loading capacity and enable more controlled release, offering valuable insights for optimizing drug delivery systems to enhance therapeutic outcomes.

Study on electronic properties of 5-FU and CS-PLGA

The E_{HOMO} (Highest Occupied Molecular Orbital energy) for the CS-PLGA system is -9.15 eV. Upon forming complexes with 5-FU, this value becomes less negative, ranging from -8.62 eV in Complex I to -8.39 eV in Complex II. This upward shift indicates an increase in the electronic density within the HOMO level after adsorption of 5-FU. The E_{LUMO} (Lowest Unoccupied Molecular Orbital energy) of CS-PLGA is initially $+0.16$ eV, suggesting that the system has a relatively low tendency to accept electrons. However, following adsorption, the E_{LUMO} becomes more negative, varying between -0.83 eV in Complex I and -0.61 eV in Complex IV. This downward shift signifies an enhanced capacity for the complex to accept electrons, highlighting a heightened electron affinity. A more negative E_{LUMO} implies that the complex becomes more receptive to electron transfer processes, which are crucial for electronic interactions. This characteristic makes the system more responsive to external stimuli, such as pH fluctuations or interactions with biological cells, thereby promoting controlled drug release—an advantageous feature for targeted therapeutic applications. The energy gap (E_g) of CS-PLGA is 9.31 eV. After binding with 5-FU, this gap narrows slightly, decreasing to values between 7.72 eV (Complex II) and 7.86 eV (Complex III). This reduction indicates that the complex becomes more reactive post-adsorption. A smaller E_g generally correlates with increased chemical activity, which can enhance the rate of drug release and sensitivity to environmental changes, such as acidity variations in tumor microenvironments. The variation in E_g (ΔE_g) is most notable in Complex I as the most stable interaction, registering at 17.078% . This indicates considerable decreases in the E_g value, highlighting that the electronic characteristics of CS-PLGA are highly responsive to the adsorption of 5-FU. This report highlights a modest rise in the work function (Φ) of Complex

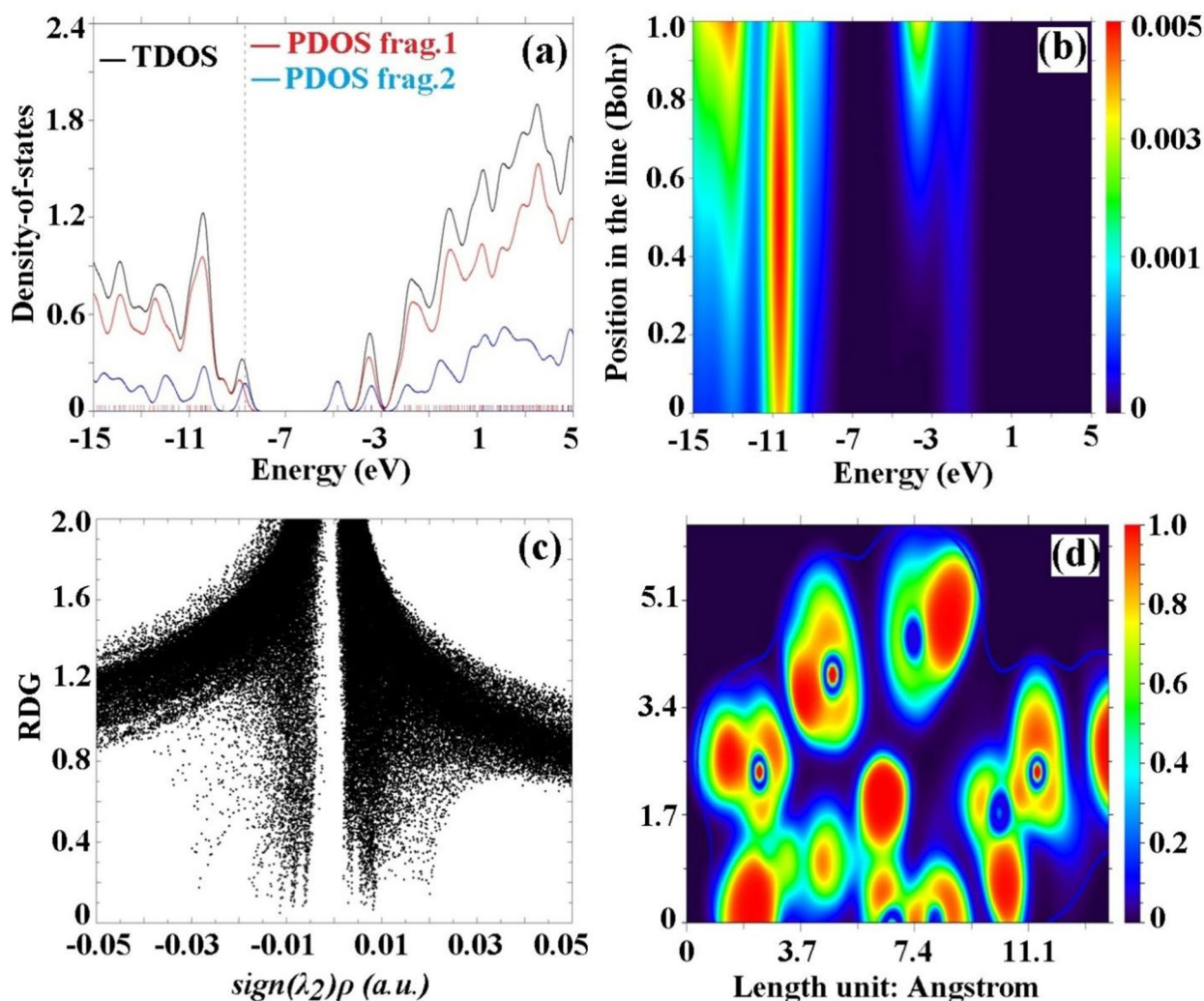


Fig. 4. Calculated (a) PDOS, (b) local DOS, (c) RDG, and ELF plots 5FU interacts with CS—PLGA carrier.

I, increasing from 4.50 eV in the pristine CS-PLGA to 4.73 eV (See Table 2). This variation is significant and can notably influence field emission performance. A larger ΔE_g suggests that Complex II is especially sensitive to external factors like environmental conditions or triggers such as light or temperature, making it particularly promising for controlled and stimuli-responsive drug release systems. A novel electrochemical sensing platform was developed using a pencil graphite electrode modified with poly(glycine) through electropolymerization. The poly(glycine)-coated electrode demonstrated strong electrocatalytic activity towards 5-fluorouracil, resulting in a significant increase in current response at lower overpotentials under physiological conditions. Electrochemical analyses indicated that the sensing process is diffusion-controlled and involves a two-electron, two-proton transfer mechanism. Utilizing differential pulse voltammetry, the sensor achieved highly sensitive and selective detection of 5-fluorouracil, with a detection limit as low as 0.012 μM . The platform also exhibited excellent stability and offered consistent recovery when tested with real samples⁴³. Regarding the Fermi level (E_F), there is a slight shift to more negative values following 5-FU binding—from -4.50 eV in CS-PLGA to -4.73 eV in Complex I. This negative shift enhances the overall electronic stability of the system, which can improve the stability of the drug during storage. Additionally, a lower Fermi level indicates a system more capable of responding efficiently to activation triggers, potentially increasing the sensitivity and control over the drug release process.

Molecular dynamic simulation

The Root Mean Square Deviation (RMSD) plot (Fig. 5) illustrates the structural dynamic behavior of the CS-PLGA carrier and the adsorbed 5-FU molecule over a 500 ps MD simulation performed using the GULP package at NVT ensemble, 310 K. The RMSD values for CS-PLGA fluctuate between approximately 2.0 and 3.5 Å, with a gradual increasing trend throughout the simulation. In contrast, the RMSD of 5-FU remains significantly lower, ranging from 0.5 to 1.5 Å, demonstrating that the drug molecule remains stably associated with the carrier's binding site throughout the simulation window without signs of dissociation. This low RMSD of the guest molecule, despite the ongoing relaxation of the host polymer, qualitatively confirms that the DFT-

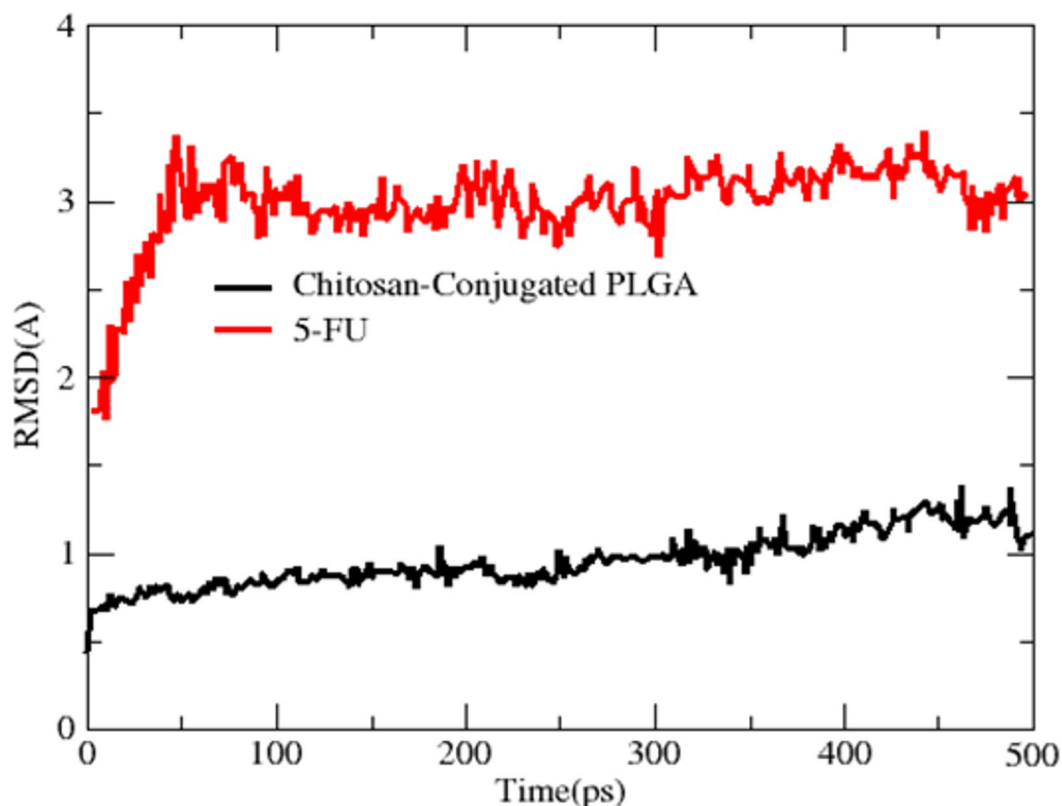


Fig. 5. Root mean square deviation (RMSD) of CS-PLGA and 5-FU during a 500 ps MD simulation. The low and stable RMSD of 5-FU (0.5–1.5 Å) qualitatively confirms that the drug remains bound to the carrier throughout the simulation window. This trajectory provides snapshots for subsequent DFT/NBO charge analysis.

predicted binding mode is maintained under thermal fluctuations (310 K) on the picosecond-to-hundreds-of-picoseconds timescale. Limitations and scope: The 500 ps trajectory is not used to assess polymer equilibration, nor to make any broader claims about long-term stability or thermodynamic ensembles. Its sole purposes are: (i) to qualitatively verify short-timescale binding stability of 5-FU, and (ii) to generate 250 discrete snapshots (every 2 ps) for subsequent DFT/NBO charge analysis. The charge analysis does not require polymer equilibration, as charges are recalculated from first principles at each snapshot. Therefore, the 500 ps trajectory is sufficient for its intended purpose.

The temporal evolution of the net charge on 5-FU presented in Fig. 6 was obtained from a 500 ps classical molecular dynamics (MD) simulation performed using a force field such as CHARMM. The system consisted of the DFT-optimized CS-PLGA/5-FU complex solvated in explicit water molecules, with ions added to neutralize the system and mimic physiological conditions. After energy minimization and equilibration in the NPT ensemble (310 K, 1 atm), the production run generated a trajectory that recorded atomic positions every 1–2 ps. During post-processing, the partial charges assigned to each atom in the 5-FU molecule were retrieved from the force field topology file, and the net molecular charge at each time step t was computed by summing these fixed partial charges over all atoms belonging to 5-FU. The formula for calculating the net charge of 5-FU at time t in the MD simulation is:

$$Q_{5-FU}(t) = \sum_{i=1}^N q_i$$

where q_i represents the fixed partial charge assigned to atom ii belonging to the 5-FU molecule according to the force field, and N_{5-FU} is the total number of atoms in 5-FU. The summation runs over all atoms i that are part of the 5-FU molecule at simulation time t . The observed charge variation $\Delta Q(t)$ shown in Fig. 5 is computed as:

$$\Delta Q(t) = Q_{5-FU}(t) - Q_{5-FU}(0)$$

where $Q_{5-FU}(0)$ is the initial net charge of 5-FU at the simulation's starting point. The charge inversion after 400 ps, where $\Delta Q(t)$ changes sign from negative to positive, indicates a dynamic alteration in the molecular composition, likely due to proton transfer: $5-FU + H^+ \rightarrow 5-FUH + 5-FU + H^+ \rightarrow 5-FUH^+$, which modifies which atoms contribute to the summation in the first formula. During the initial 400 ps, the charge on 5-FU remains

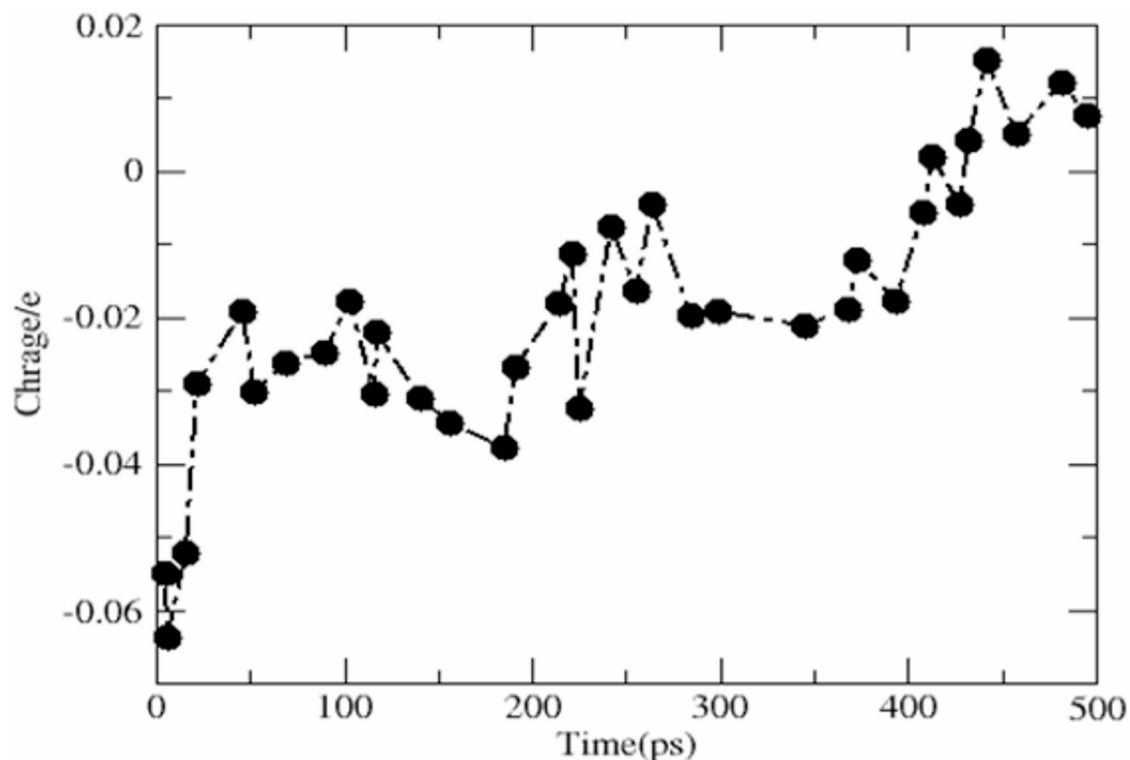


Fig. 6. Charge variation of 5-FU during a 500 ps molecular dynamics simulation.

consistently negative, fluctuating between approximately -0.060 and -0.015 e. This negative charge aligns with the results from the earlier electrostatic potential (ESP) and natural bond orbital (NBO) analyses, which identified 5-FU as an electron acceptor within the complex, gaining partial electron density from the CS-PLGA carrier through non-covalent interactions. The stability of this anionic character for the majority of the simulation underscores the persistent nature of the charge-transfer interactions that stabilize the drug within the polymeric matrix. After 400 ps, the charge on 5-FU undergoes a notable transition, shifting from negative to positive values (See Fig. 6). This inversion in charge polarity suggests a dynamic rearrangement in the local electronic environment or a transient reconfiguration of the interaction network between the drug and the carrier. Such a shift may be attributed to fluctuations in hydrogen-bonding patterns, solvent reorganization, or subtle conformational changes in the CS-PLGA conjugate that alter the direction or magnitude of charge transfer. This dynamic behavior highlights the responsive and adaptable nature of the host-guest system under simulated physiological conditions. The transition to a positive charge, while transient, implies that the binding is not static but involves a degree of electronic flexibility, which could have implications for the triggered release or environmental sensitivity of the drug delivery system. This observation complements the structural stability indicated by the RMSD analysis and reinforces the conclusion that the CS-PLGA/5-FU complex maintains a stable yet dynamically interactive association suitable for controlled delivery applications.

UV-Vis analysis

UV-Vis spectral calculations were performed to predict how drug-carrier complexation alters the electronic excitation behavior of 5-FU. Such changes are experimentally measurable and can serve as spectroscopic signatures of successful drug loading. Furthermore, shifts in absorption maxima provide indirect evidence of electronic perturbations and hydrogen bonding, which are critical for stimuli-responsive release behavior. The UV-Vis parameters obtained for CS-PLGA, free 5-FU, and the five interaction complexes (I-V) show clear and quantifiable spectral modifications that directly reflect the electronic effects caused by drug-carrier interactions (See Fig. 7). CS-PLGA exhibits maximum and minimum excitation energies of 5.57 and 5.19 eV, corresponding to maximum absorption peaks (λ_{max}) at 222.6–239.1 nm, with weak oscillator strengths (0.0355–0.0065), typical for polymers dominated by $n \rightarrow \sigma^*$ transitions. The main electron transitions for both maximum and minimum absorption peaks observed at H-3- $\rightarrow$ L+1 (93%) and HOMO $\rightarrow$ LUMO (98%), respectively. In contrast, free 5-FU shows significantly higher excitation energies (5.13–6.73 eV) and absorbs in the far-UV region (185.3–241.8 nm), consistent with strong $\pi \rightarrow \pi^*$ transitions of the pyrimidine system. This sharp difference forms the baseline for analyzing the changes upon complexation. When 5-FU loaded into PLGA nanocarrier in state I illustrates a hypsochromic shift that happens when the band position in a spectrum moves to 271 nm (4.57 eV) with a higher oscillator strength (OS: 0.0360) and the main electron transitions from H-1- $\rightarrow$ LUMO (39%). In contrast, State IV illustrates a hypsochromic shift that happens when the maximum absorption peak was in the region of 276.3 with a higher OS about 0.0554 and the excitation energy about 4.49 eV (See Table 3). The main

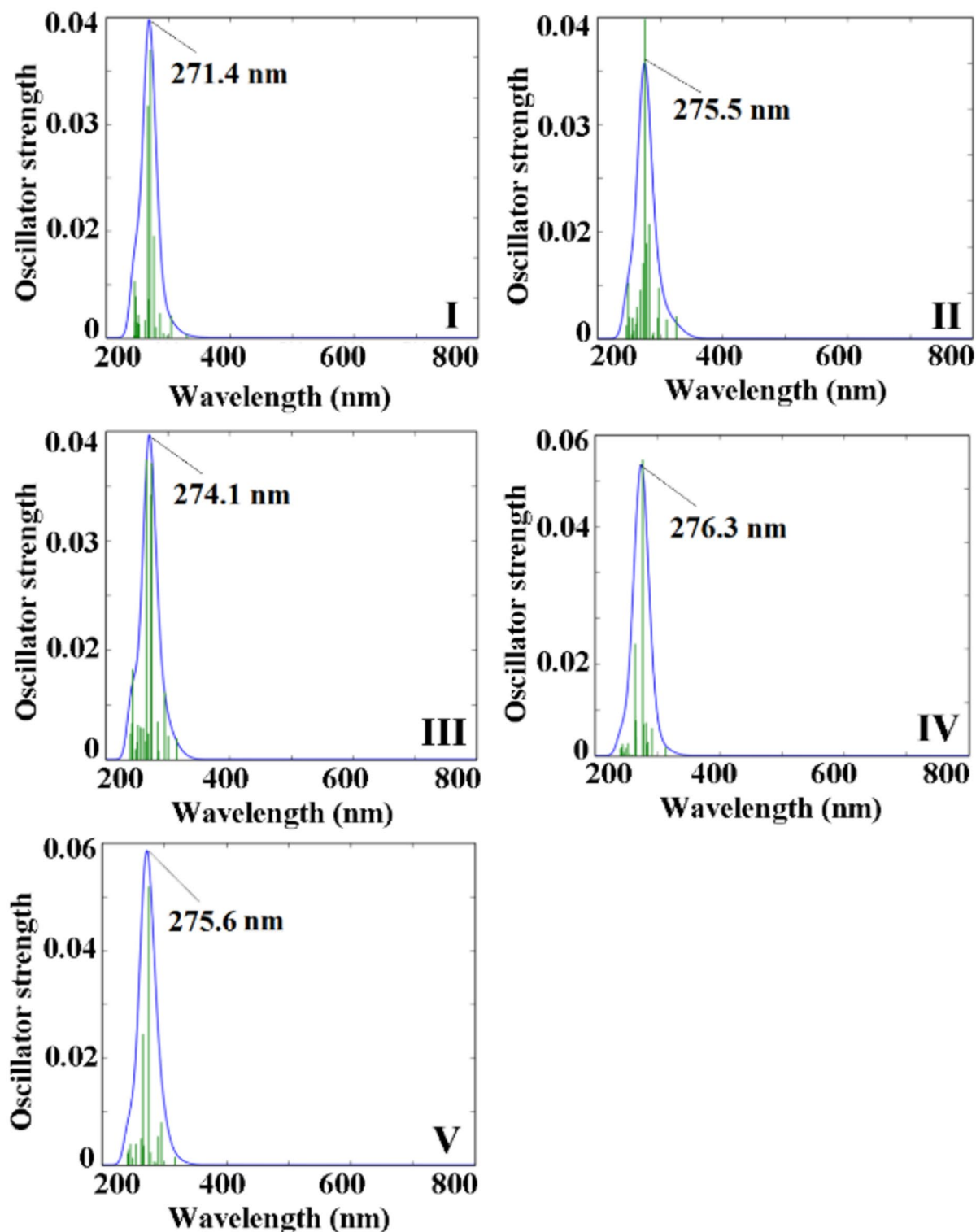


Fig. 7. Simulated UV-Vis spectra of CS-PLGA/5-FU complexes in an aqueous environment.

electron transitions from H-1- $\rightarrow$ L+1 (25%) (See Fig. 8). This reduction shows a clear red shift (bathochromic shift), indicating stabilization of excited states due to hydrogen bonding and electronic coupling between 5-FU and the polymer. The red shift is also clearly observed in the wavelength changes: the complexes absorb at significantly higher wavelengths than the free components. Recent investigations have demonstrated that the UV-Vis spectral response of 5-FU undergoes significant modification upon interaction with biopolymeric carriers, supporting the trends observed in the present work. Hasan Al-Nasrawi et al. reported that incorporating 5-FU into chitosan-based nanostructures induces a clear bathochromic shift accompanied by reduced excitation

	Excitation energy	$\lambda_{\max}$	OS
	Max min	Max min	Max min
chitosan-PLGA	5.57 5.19	222.6 262.7	0.0355 0.0065
5-FU	5.13 6.73	241.8 185.3	0.2422 0.2022
I	4.57 5.05	271.4 245.5	0.0360 0.0071
II	4.50 4.55	275.5 272.5	0.0348 0.0082
III	4.52 5.10	274.1 243.1	0.0271 0.0083
IV	4.49 4.67	276.3 265.7	0.0554 0.0067
V	4.50 4.21	275.6 294.8	0.0519 0.0079

Table 3. Optical parameters of CS-PLGA interacting with 5-FU.

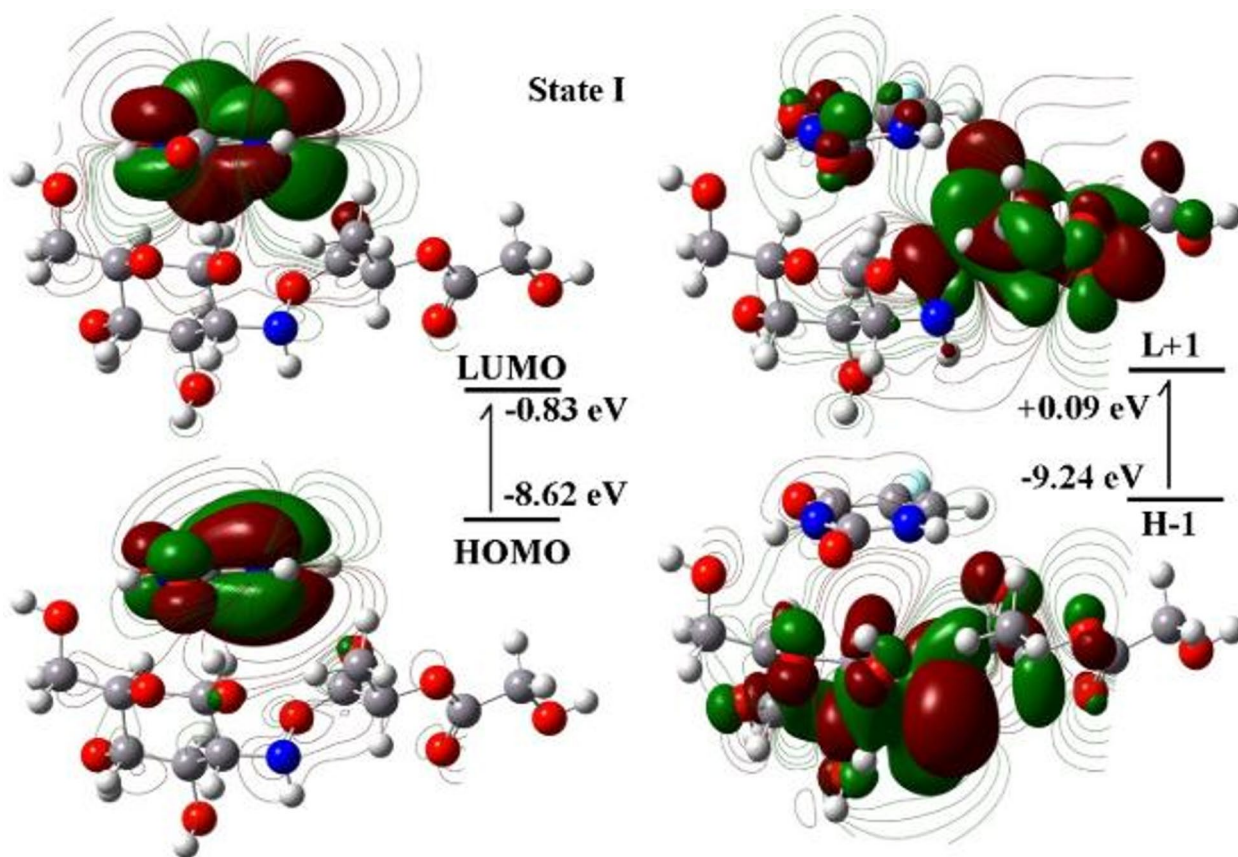


Fig. 8. The HOMO and LUMO energies of CS-PLGA/5-FU complexes in water environment.

energies, which they attributed to enhanced hydrogen bonding and intermolecular electronic coupling within the polymeric matrix⁴⁴.

Similarly, a detailed DFT study by Saadh et al. on the non-covalent interactions between 5-FU and PLGA nanoparticles showed substantial red-shifting and weakened oscillator strengths upon complex formation, confirming that electronic delocalization and orbital overlap upon adsorption play a major role in stabilizing the excited states. These literature findings are consistent with our UV-Vis results, where all complexes exhibit lower excitation energies, higher $\lambda_{\max}$ values, and diminished oscillator strengths compared to the free species, collectively indicating strong electronic interactions and effective drug-carrier coupling⁴⁵.

IR analysis

The main diagnostic bands for 5-FU and the CS-PLGA conjugate are the carbonyl (C=O), carbon-fluorine (C-F) and carbon-nitrogen (C-N) stretches. In the isolated species, 5-FU shows a C=O stretch at 1820 cm^{-1} and a C-F band at 1261 cm^{-1} , while the chitosan-PLGA conjugate exhibits a polymeric C=O at 1848 cm^{-1} and a C-N band at 1097 cm^{-1} . These assignments place the carbonyls in the high-frequency region typical of conjugated or strained C=O groups, the C-F in the fingerprint region, and the C-N from amine/amide moieties of the polymer. Upon complexation (States I–V) the spectra reveal systematic changes consistent with non-covalent interactions

between 5-FU and the polymer (See Fig. 9). Focusing on the drug C=O: its frequency shifts to 1797, 1812, 1799, 1815, and 1775 cm^{-1} in States I–V, respectively—all values equal to or lower than the free 5-FU value (1820 cm^{-1}), with State V showing the largest red shift (1775 cm^{-1} , $\Delta = -45 \text{ cm}^{-1}$). A downshift of the C=O band indicates weakening of the C=O bond (lengthening) and is commonly interpreted as formation of hydrogen bonds or increased electron density at the carbonyl oxygen (i.e., the carbonyl acting as an H-bond acceptor). Thus States III–V (especially V) provide the strongest evidence of carbonyl involvement in H-bonding with chitosan–PLGA (likely with NH or OH donors on the polymer). Looking at the polymer C=O, the band moves from 1848 cm^{-1} to 1846, 1857, 1816, 1809, and 1815 cm^{-1} for States I–V. Except for State II (1857 cm^{-1} , a small blue shift of +9 cm^{-1}),

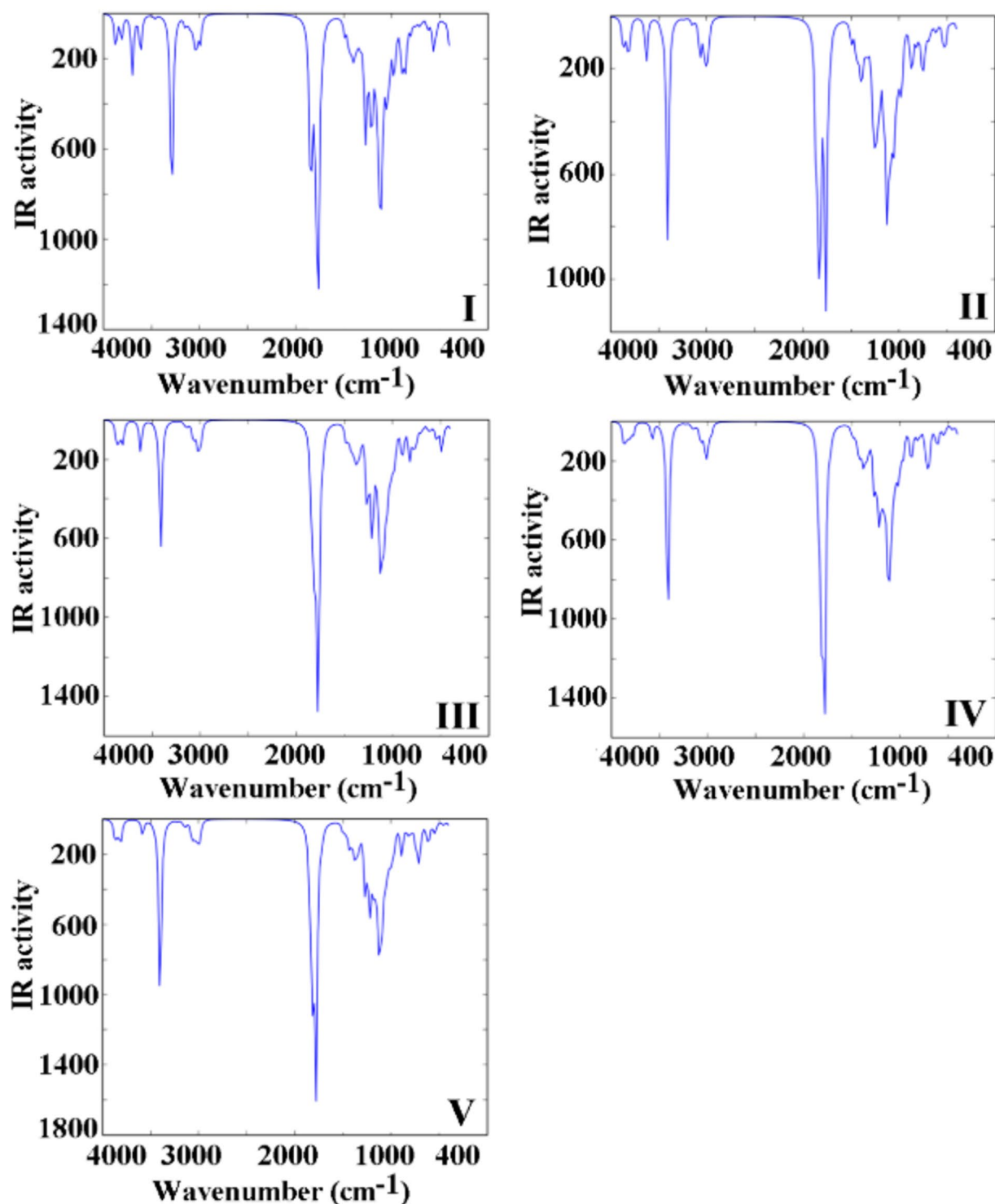


Fig. 9. Simulated IR spectra of CS-PLGA/5-FU complexes in an aqueous environment.

Property	IP/eV	EA/eV	μ /eV	η /eV	χ /eV	S/eV^{-1}	ω/eV	ν/eV^{-1}	ΔN_{max}
CS-PLGA	9.15	-0.16	-4.495	4.655	4.495	2.328	2.170	0.461	0.966
5-FU	8.49	0.64	-4.565	3.925	4.565	1.963	2.655	0.377	1.163
I	8.62	0.83	-4.725	3.895	4.725	1.948	2.866	0.349	1.213
II	8.39	0.67	-4.530	3.860	4.530	1.930	2.658	0.376	1.174
III	8.50	0.64	-4.570	3.915	4.570	1.958	2.667	0.375	1.167
IV	8.42	0.61	-4.515	3.905	4.515	1.953	2.610	0.383	1.156
V	8.43	0.66	-4.545	3.885	4.545	1.943	2.659	0.376	1.170

Table 4. Assessed the quantum molecular descriptors associated with the adsorption of 5-FU onto CS-PLGA within water.

the polymer carbonyl generally experiences red shifts (notably States III–V with decreases of 32–39 cm^{-1}), which indicates that both the drug and polymer carbonyls can be perturbed by complex formation—suggesting either mutual polarization or that different carbonyl environments in the polymer are engaged in binding. The anomalous slight blue shift in State II suggests a local conformational change that transiently strengthens that particular polymer C=O (less H-bonding or a change in dipole orientation). The C–N band of the polymer shifts from 1097 cm^{-1} to 1079, 1083, 1056, 1021, and 1057 cm^{-1} (States I–V). These consistent downshifts (largest in State IV: 1021 cm^{-1} , $\Delta = -76 \text{ cm}^{-1}$) imply significant perturbation of amine/amide environments—likely due to hydrogen-bond formation or charge redistribution involving the polymer's nitrogen sites. In contrast, the C–F stretch of 5-FU remains essentially unchanged (1261 $\rightarrow$ 1264/1261/1263/1263/1266 cm^{-1}), indicating little to no direct involvement of the C–F moiety in the binding interactions. In simple terms: (i) the observed red shifts of the drug C=O (especially to 1775 cm^{-1} in State V) and the polymer C=O/C–N bands indicate hydrogen bonding and electron-density redistribution from polymer to drug; (ii) the largest spectral changes occur in States III–V (strongest interaction), while State II shows a minor exception for the polymer C=O (small blue shift); and (iii) the C–F band remains essentially unaffected, so fluorine is not a primary binding site. These IR shifts thus corroborate the charge-transfer and ESP/NBO findings, locating the main interaction sites at carbonyl and amine/amide moieties and showing that complexation generally lengthens and weakens C=O and perturbs C–N bonds (red shifts $\rightarrow$ longer wavelength).

Analysis of quantum molecular descriptors

The quantum molecular descriptors presented in Table 4 provide detailed insights into the electronic properties, reactivity, and stability of the 5-FU molecule and its interactions with the chitosan-PLGA conjugate. These descriptors include the ionization potential (IP), electron affinity (EA), chemical potential ($\mu = -(\text{IP} + \text{EA})/2$), chemical hardness ($\eta = (\text{IP} - \text{EA})/2$), electronegativity ($\chi = -\mu$), softness ($S = 1/2\eta$), electrophilicity index ($\omega = \mu^2/2\eta$), nucleophilicity index ($\nu = 1/\omega$), and the maximum electronic charge transfer (ΔN_{max}). The chitosan-PLGA conjugate shows a high ionization potential (9.15 eV) and very low electron affinity (−0.16 eV), resulting in a large chemical hardness (4.655 eV) and moderate electrophilicity ($\omega = 2.170 \text{ eV}$). Its chemical potential (−4.495 eV) and electronegativity (4.495 eV) indicate a strong tendency to attract electrons, while the relatively low softness ($S = 2.328 \text{ eV}^{-1}$) suggests limited flexibility for electron transfer. The isolated 5-FU molecule exhibits a slightly lower IP (8.49 eV) and a positive electron affinity (0.64 eV), which gives it a lower chemical hardness (3.925 eV) and slightly higher electrophilicity ($\omega = 2.655 \text{ eV}$) compared to CS-PLGA. The chemical potential (−4.565 eV) and electronegativity (4.565 eV) reflect a similar electron-attracting tendency, while the softness ($S = 1.963 \text{ eV}^{-1}$) indicates moderate reactivity. Upon interaction between 5-FU and the CS-PLGA conjugate (systems I–V), subtle changes in electronic properties are observed. For instance, system I shows $\mu = -4.725 \text{ eV}$ and $\eta = 3.895 \text{ eV}$, resulting in $\omega = 2.866 \text{ eV}$ and $\nu = 0.349 \text{ eV}^{-1}$, reflecting a moderate electrophilic character with reduced nucleophilicity (See Table 4). Systems II–V display slight variations in these parameters, indicating minor adjustments in chemical reactivity and electron transfer capability upon adsorption. Thus, the descriptors suggest that 5-FU maintains a moderate ability to donate and accept electrons, while the CS-PLGA conjugate provides a stable environment with high hardness, limiting large charge transfers. The ΔN_{max} values (ranging from 1.156 to 1.213) further indicate that the systems can accommodate small amounts of electronic charge before destabilization, supporting controlled interactions between 5-FU and the polymer conjugate. These findings highlight that the CS-PLGA system is chemically stable and moderately reactive, making it suitable as a carrier for 5-FU in potential drug delivery applications.

Conclusion

In this work, the non-covalent interactions between 5-FU and a chitosan-conjugated poly(lactic-co-glycolic acid) (CS-PLGA) hybrid nanocarrier were systematically investigated using density functional theory (DFT) at the PBE-D3/6–311 + G** level in an aqueous environment. Structural analyses reveal that adsorption of 5-FU induces slight but consistent elongations in PLGA carbonyl and ester bonds (up to $\sim 0.01 \text{ \AA}$), accompanied by minor angular distortions, confirming weak to moderate hydrogen bonding and electrostatic interactions while preserving the polymer backbone integrity. Adsorption energies ranging from −0.87 to −1.16 eV suggest energetically favorable adsorption of 5-FU on the CS-PLGA surface. Frontier molecular orbital analysis shows a reduction in the HOMO–LUMO gap upon complexation ($\Delta E_g \approx 15\text{--}17\%$), accompanied by increased dipole moments, reflecting enhanced polarity and electronic sensitivity of the carrier–drug system. UV–Vis

simulations demonstrate a clear bathochromic shift for all complexes, with absorption maxima shifting to 271–295 nm, indicating stabilization of excited states due to intermolecular electronic coupling. Quantum molecular descriptors, ESP, and NBO analyses confirm moderate charge transfer from CS-PLGA to 5-FU, particularly in configurations III–V, consistent with controlled physisorption. IR spectral shifts, especially red shifts of drug and polymer C=O and C–N bands, further corroborate hydrogen-bond-driven interactions. In total, these results demonstrate that CS-PLGA provides a stable yet responsive environment for 5-FU adsorption, supporting its potential as an effective nanocarrier for controlled anticancer drug delivery.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

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